

variable `t1`. This is due to the presence of the address of operator `&` that precedes the identifier `a1`. Thus a pointer may be defined as a variable that stores the reference to another variable.

Pointers are special variables that store the reference of a variable of its type. That means an integer pointer can store the address of an integer variable, a float pointer can store the address of a float variable and so on. In the above examples, all variables used to store reference of other variables are pointers. Pointers play a vital role and have uses in various applications. Generally a pointer is defined as follows:

```
type * name;
```

Here, `type` refers to the data type of the variable that the pointer points to. Let us take another example that is shown below.

```
int * n1;  
char * c1;  
float * g1;
```

Here, three pointers have been declared. Each pointer that is declared points to a different data type, but all of them occupy the same amount of space in the memory. However, the data to which the pointers point to, are of different types - the first pointer points to an `int`, the second points to a `char` and last one points to a `float`. Now let us look at a program which is based on pointers.

```
// Program 1  
// Application of pointers  
#include <iostream.h>  
int main ()  
{  
    int f1, s1;  
    int * m1;
```